

Backend Schema Document

Pipeline Payload

Nodes

```
{
  "id": "node-1",
  "type": "inputNode",
  "position": {
    "x": 100,
    "y": 100
  },
  "data": {}
}
```

Edges

```
{
  "id": "edge-1",
  "source": "node-1",
  "target": "node-2"
}
```

Internal Graph Structure

Adjacency List

```
graph = {
  "node1": ["node2"],
  "node2": ["node3"]
}
```

Validation Objects

PipelineRequest

```
nodes: List
edges: List
```

PipelineResponse

```
num_nodes: int
num_edges: int
is_dag: bool
```

Relationships

Node

1 → Many Edges

Edge

Many → 1 Source Node

Many → 1 Target Node

Validation Rules

Node IDs must be unique

Edge source must exist

Edge target must exist

No malformed edges

DAG Rules

Cycle present:

False

No cycle:

True

Ownership

Single-session ownership

No user management required

No authentication required